

IFRS9 Vasicek Models: From Segment to Rating Grade

Challenges in Point-in-Time Calibration Across Multiple Levels of Granularity

Andrija Djurovic

www.linkedin.com/in/andrija-djurovic

The Vasicek PD Model in IFRS9 Modeling

- The Vasicek Probability of Default (PD) model is one of the most commonly used approaches for PD modeling under IFRS9.
- Typically, banks use the risk differentiation functions of Internal Ratings-Based (IRB) models within the Vasicek framework to produce IFRS9-required Point-in-Time (PIT) estimates at the rating-grade level.
- The calibration of PIT estimates is closely linked to the Forward-Looking Information (FLI) models used in the IFRS9 framework.
- Although different designs and implementations of the Vasicek model can be observed in practice, one of the most common approaches relies on extracting the systemic factor from observed default rates at the segment level. This factor is then linked to the macroeconomic environment through the FLI model, and its forecasts are used to translate segment-level forecasts into shifts at the rating scale level.
- Moving from segment-level forecasts of the systemic factor to their application at the rating scale level creates several challenges for practitioners. One of the most apparent challenges is the potential mismatch between the observed default rates at the segment level and the fitted PIT estimates at the rating grade level.

The Vasicek PD Model in IFRS9 Modeling cont.

- The existence of this mismatch raises the question of how this characteristic of the Vasicek model should be interpreted and whether it poses a significant risk for model implementation and backtesting.
- The purpose of this presentation is to demonstrate this mismatch and to raise awareness of the consequences of modeling choices during model development and validation. It should be noted that this potential mismatch is not unique to the Vasicek model and may also arise in other approaches used for the same purpose. Therefore, practitioners are encouraged to build upon the presented work and to test and compare different implementations and designs of the Vasicek model, as well as alternative approaches for IFRS9 PD modeling.
- The remainder of this presentation is organized as follows: the next two slides present the foundations of the Vasicek model and the Maximum Likelihood Estimation (MLE) method used to estimate the parameters of the Vasicek distribution. The remaining slides describe the simulation study design and present the corresponding results, highlighting the differences between the observed and model-fitted default rates.

The Vasicek Distribution

- Essentially, the Vasicek model relies on the assumption that observed default rates follow the Vasicek distribution.
- Vasicek distribution is a two-parameter ($0 < p < 1$ and $0 < \rho < 1$) continuous distribution on the range 0 to 1. If a variable x has a Vasicek distribution, then x (observed default rate) can be represented as:

$$x = \Phi \left(\frac{\Phi^{-1}(p) - \sqrt{\rho}Z}{\sqrt{1 - \rho}} \right)$$

where:

- p and ρ are the parameters of the distribution, commonly referred to as the average default rate and asset correlation, respectively;
 - Z represents the systemic factor drawn from the standard normal distribution;
 - Φ and Φ^{-1} denote the distribution and quantile function of the standard normal distribution, respectively.
- Given the parameters p and ρ , which are commonly estimated at the segment (portfolio) level, the systemic factor Z is derived by manipulating the equation of the Vasicek distribution.

The MLE of the Vasicek Distribution Parameters

- It should be noted that several approaches are available for estimating the parameters of the Vasicek distribution. However, the Maximum Likelihood Estimation (MLE) method generally produces the least biased estimates of the asset correlation parameter (ρ) and, therefore, is used in the simulation study presented herein.
- In essence, the MLE method is based on maximizing the log-likelihood function of the Vasicek Probability Density Function (PDF).

The MLE of the Vasicek Distribution Parameters cont.

- The Log-Likelihood of the Vasicek PDF is given by:

$$\sum_{i=1}^T \ln(f_{p,\rho}(x_i))$$

with

$$f_{p,\rho}(x) = \sqrt{\frac{1-\rho}{\rho}} e^{\frac{1}{2} \left(\phi^{-1}(x)^2 - \left(\frac{\sqrt{1-\rho}\phi^{-1}(x) - \phi^{-1}(\rho)}{\sqrt{\rho}} \right)^2 \right)}$$

where:

- T represents the number of observations;
 - $f_{p,\rho}(x)$ denotes the Vasicek Probability Density Function;
 - x represents the observed default rates;
 - p denotes the average default rate; and
 - ϕ^{-1} is the quantile function of the standard normal distribution.
- With p calculated as $\frac{\sum_{i=1}^T x_i}{T}$, ρ is derived by maximizing the Log-Likelihood function based on the observed default rates.

Simulation Setup

The purpose of this simulation is to demonstrate the mismatch between segment- and rating grade-level default rates in one of the most common implementations of the Vasicek model in IFRS9 PD modeling. Specifically, this occurs when the systemic factor (Z) is calculated at the segment level and subsequently applied at the rating scale level to produce fitted PIT estimates. As a result, even for observed reference periods, discrepancies between observed and fitted segment-level default rates can be expected.

The dataset used in this simulation is available at the following [link](#). The steps below outline the simulation framework:

- 1 Based on the transition rates for each reference period (Period) from 1 to 20, the number of obligors per rating grade (Obs), and the default rates (D), calculate the segment-level default rates.
- 2 Using the time series of observed default rates obtained in step 1, calibrate the parameters of the Vasicek distribution (p and ρ) using the MLE method.
- 3 After calibrating the parameters in step 2, calculate the systemic factor Z for each Period.
- 4 Use the calculated systemic factor Z , together with the TTC transition matrix and the estimated parameters of the Vasicek distribution, to derive fitted PIT estimates at the rating grade level.
- 5 Recalculate the segment-level default rates for each Period based on the fitted PIT estimates.
- 6 Compare the observed and fitted default rates for each Period at both the segment and rating grade levels.

Note

Practitioners should note that the dataset used in this simulation was created solely for demonstration purposes; therefore, the observed differences may vary when applied to real-world datasets. Furthermore, the simulation framework can be readily extended to compare alternative implementation approaches for the Vasicek model and other methodologies for IFRS9 PD modeling. Practitioners are encouraged to further explore and build upon the ideas presented here.

Simulation Results

The following tables present the intermediate results required as inputs for the calculation of the fitted PIT estimates at the segment and rating scale levels.

Segment ODR

```
## [1] 0.0299 0.0526 0.0507 0.0482 0.0371 0.0509 0.0487 0.0318 0.0226 0.0366
## [11] 0.0346 0.0376 0.0283 0.0430 0.0439 0.0318 0.0302 0.0455 0.0445 0.0286
```

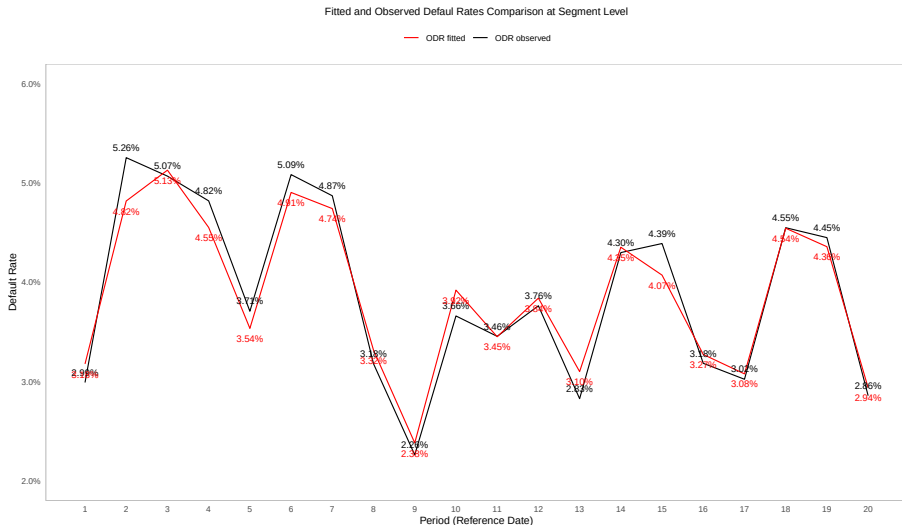
MLE Estimates of the Vasicek Model Parameters

```
## method      p      rho
## 1      MLE 0.0389 0.0115
```

Systemic Factor Z - Segment Level

```
## [1] -1.0013  1.4282  1.2648  1.0385 -0.1041  1.2785  1.0857 -0.7495 -2.1185
## [10] -0.1587 -0.4022 -0.0413 -1.2294  0.5331  0.6251 -0.7433 -0.9567  0.7820
## [19]  0.6842 -1.1863
```


Simulation Results cont.



Simulation Results cont.

Fitted and Observed Default Rates Comparison at Rating Grade Level

